



ID:

1. Convert **last two digits of your student ID** to a **7 bit binary** number:

[1]

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Your ID (XY)

Binary Representation of Your ID

2. A communication channel operates with a bandwidth of **XY kHz** and has a signal-to-noise ratio (SNR_{dB}) of **20 dB**. Determine the **maximum theoretical data rate** that can be achieved. (XY = last two digits of your student ID)

[5]

3. Convert **binary representation of student ID** into digital signals using the following schemes.

(i) Manchester

(ii) AMI

[8]

4. You want to send a message to your friend Ashton Hall, who is 13,247 km away. Each letter in the message consists of **8 bits** of data. The signal propagates through the medium at a speed of $2.8 * 10^8$ m/s, and the data transmission speed of your network is **XY Kbps (kilo bits per second, XY is the last two digits of your student ID)**. The message you want to send is:

SARATOGA

Calculate the **transmission** time.

[3]

5. **(Mandatory)** Write one good thing and two suggestions for improvement about your course teacher.

[3]